

Uncoupling myelin membrane composition from tissue mechanics: consequences of Schwann-cell-specific LXR β deletion

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Abstract:

The myelin sheath of the peripheral nervous system is among the most lipid-rich biomembranes in the body: ~70% of its dry mass is lipid, dominated by cholesterol and sphingolipids, and organized into a compact multilamellar structure through the repeated wrapping of Schwann cell plasma membranes. How the lipid composition of this membrane is sensed and maintained, and how composition couples to the mechanical integrity of the nerve, remain open questions at the interface of membrane biophysics and cell biology.

Liver X Receptor β (LXR β) is a cholesterol-oxysterol-sensing nuclear receptor that governs cholesterol homeostasis, lipogenesis and lipid efflux. We generated a Schwann-cell-specific LXR β knockout mouse (Dhh:Cre; LXR $\beta^{fl/fl}$) to ask how the loss of a membrane-lipid sensor reshapes the myelin membrane, its composition and its mechanics.

Schwann-cell LXR β ablation resulted in a near-complete absence of myelination from early embryonic stages, progressing to total Schwann cell loss in adulthood, accompanied by profound depletion of myelinated axons and severely impaired nerve conduction. Chemical imaging by Time-of-Flight Secondary Ion Mass Spectrometry (ToF-SIMS) revealed a dramatic collapse of nerve cholesterol, mirrored by downregulation of the biosynthetic machinery (HMGCR, SREBP2) and compensatory upregulation of the efflux effectors ABCA1 and ApoE. Strikingly, sphingomyelin was enriched rather than depleted, indicating an active remodeling of the residual membrane lipidome.

Nanomechanical mapping of intact nerve fibers by Atomic Force Microscopy (AFM) uncovered a counter-intuitive result: despite near-absent myelin, fiber elasticity (Young's modulus) was preserved. This mechanical rescue was accompanied by increased collagen deposition, increased number of endoneurial fibroblast-like cells, and infiltration of perineurial cells that structurally re-scaffold the endoneurium.

Our work positions the myelin sheath as a tractable model biomembrane whose composition is transcriptionally tuned by a lipid sensor, and reveals an unexpected uncoupling between membrane lipid content and tissue-scale mechanics — buffered by extracellular-matrix and cellular remodeling.

Suggested keywords: myelin membrane; cholesterol; sphingomyelin; LXR β ; ToF-SIMS lipid imaging; AFM nanomechanics; membrane remodeling; Schwann cells.